

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN

2025

Bachelor in Computer Applications

Course Title: **Operating Systems**

Code No: **CACS251** Semester: **IV**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. How multitasking is different from multiprocessing? Explain microkernel in detail.
3. What is critical section problem? How Peterson's solution can be used to achieve mutual exclusion?
4. What is TLB? How can it be used in paging? Explain.
5. What are advantages of distributed system over centralized system? Explain functional steps in remote procedure call with suitable diagram.
6. What is DMA? How can it be used to transfer data? Explain.
7. Explain different attributes of a file. What is hierarchical directory system?
8. What are authentication and authorization? Explain different authentication systems.

Group C

Attempt any TWO questions[2x10=20]

9. What are internal and external fragmentations? Explain how external fragmentation occurs in segmentation. Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB, and 375 KB (in order)?
10. Consider the following set of processes, with the length of the CPU burst given in milliseconds:

Process	Arrival Time	Burst Time	Priority
P1	10	6	2
P2	12	1	1
P3	33	3	2
P4	46	2	3

a) Draw Gantt charts showing the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority), and RR (quantum = 2ms).b) Compute average waiting time and average turnaround time for each of these scheduling algorithms mentioned above.c) What will happen if the quantum size is too large or too small in RR scheduling algorithm?
11. What are the four necessary conditions for deadlock to occur? Given three process A, B, and C and three resources X, Y, and Z are following events happen in the sequence as given:a) A request Xb) B request Yc) C request Zd) A request Ye) B request Zf) C request Xg) C request YAssume that requested resource should

be always be allocated to request process if it is available. Draw the resource allocation graph for the sequence and also mention whether it is deadlock? If it is how to recover the deadlock?